

ECO208 Macroeconomic Theory

Lecture 21

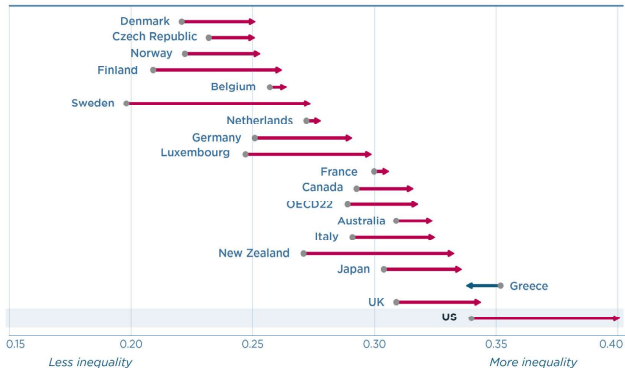
Sebastian Dyrda
University of Toronto Mississauga

UTM, April 5, 2022

Inequality before and after COVID-19.

Global Pandemic and Macroeconomic Policy
Part II

Figure 1: Change in Gini coefficient, 1985 to 2013

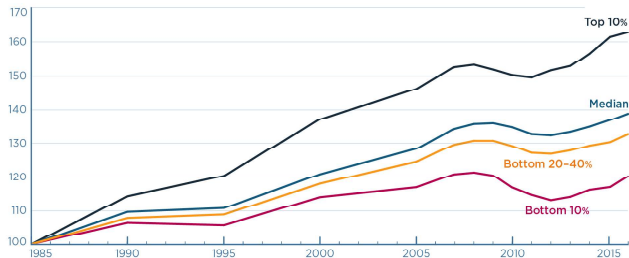


The Gini coefficient measures how equally income is distributed across a population, with 0 being perfectly equal (where everyone receives an equal share) and 1 being completely unequal (where 100 percent of income goes to only one person). Between 1985 and 2013 (or latest available year of data), the Gini coefficient rose significantly for most high-income countries for which long-time series are available.

Note: 1985 data refer to 1985 or closest available year. 2013 data refer to 2013 or nearest available year. The Gini coefficient measures how equally income is distributed across a population, from 0 (perfectly equal) to 1 (all income to one person).

Source: Organization for Economic Cooperation and Development (OECD), "In It Together: Why Less Inequality Benefits All."

Figure 2: Real disposable income in advanced economies by income position, 1985-2016 (index: 1985=100)

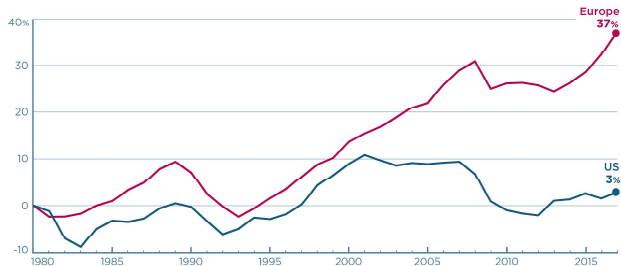


Note: Unweighted average for 17 countries for which long-term data are available: Canada, Denmark, Finland, France, Germany, Greece, Israel, Italy, Japan, Luxembourg, Mexico, Netherlands, New Zealand, Norway, Sweden, United Kingdom, and United States. Available dataset includes Mexico, which is not an advanced economy.

Source: Organization for Economic Cooperation and Development (OECD), "Under Pressure: The Squeezed Middle Class."

Disposable income (what is left after taxes and government spending) has risen the most for the top 10 percent of earners in recent decades. Poorer families have benefited much less from wider economic growth in most countries. Across many countries, middle and low incomes are barely higher or less in 2016 than what they were ten years prior.

Figure 3: Average pretax income growth of bottom 50 percent of population relative to 1980

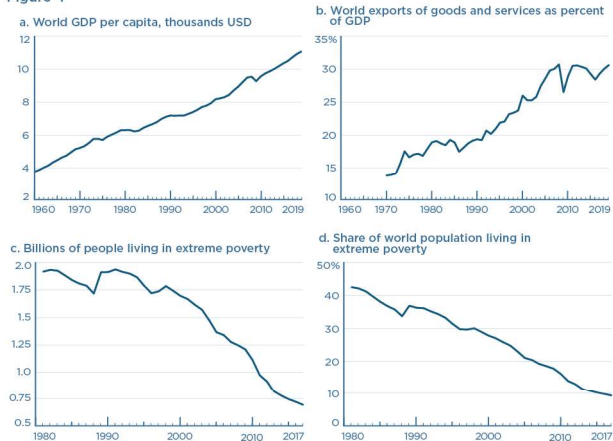


Since 1980, the pretax income of the bottom half of workers in European countries has grown by 37 percent. For the bottom half of Americans, pretax income has risen only 3 percent.

Note: Data for Europe refer to 38 European nations.

Sources: Chancel (2019) based on Blanchet, Chancel, and Gethin (2019); Piketty, Saez, and Zucman (2018).

Figure 4



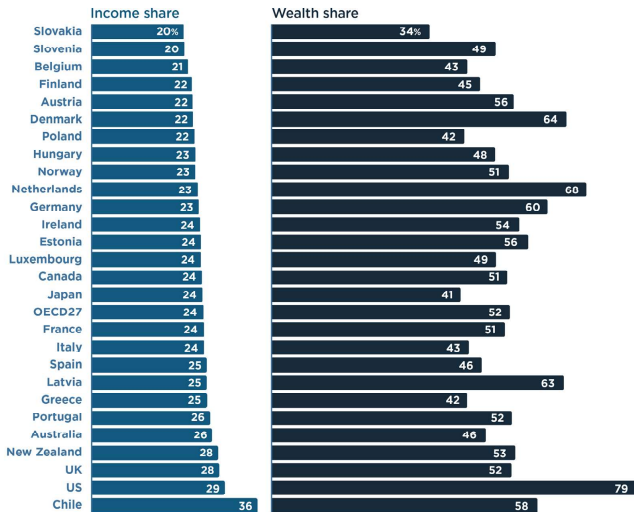
Note: GDP per capita shown in 2010 US dollars.

Source: The World Bank, "The role of trade in ending poverty."

There is a strong correlation between expanded exports as a share of global GDP and the reduction in extreme poverty, based on research by the World Bank.

Wealth inequality is higher than income inequality in advanced economies.

Figure 5: Share of total household disposable income and total household net wealth held by the top 10 percent, 2015 (or latest available)

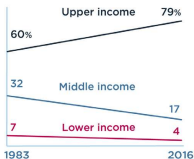


Note: The OECD average is the simple country average. Data refer to the share held by the richest 10% of households in the case of wealth; and by the richest 10% of individuals in the case of income. Numbers shown on bars are rounded.

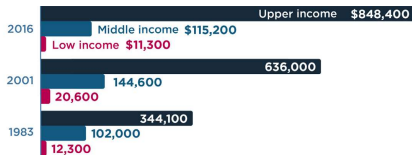
Source: Organization for Economic Cooperation and Development (OECD), "Society at a Glance 2019: OECD Social Indicators."

Figure 6

a. Share of US aggregate wealth



b. Median family wealth



Only upper-income families in the United States made lasting wealth gains in recent decades. Middle- and lower-income families lost wealth between 2001 and 2016.

Note: Families are assigned to income tiers based on their size-adjusted income. Numbers may not sum to 100 due to rounding. Median family wealth shown in 2018 US dollars.

Source: Pew Research Center, "Most Americans Say There Is Too Much Economic Inequality in the US, but Fewer Than Half Call It a Top Priority."

Figure 7: Percent of US children earning more than their parents at age 30 by year of birth, 1940-1984

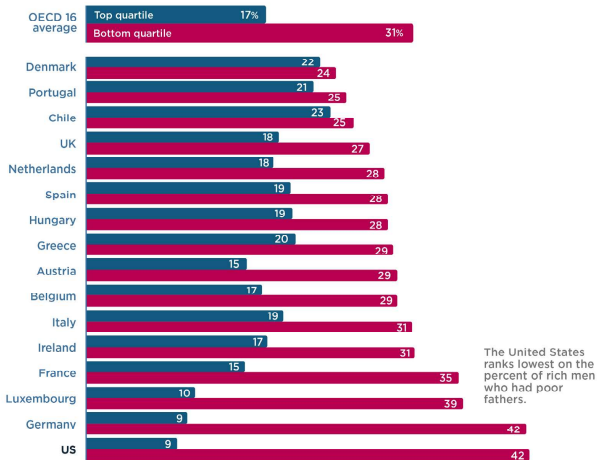


US children born in the early 1980s were much less likely to earn more at age 30 than their parents did at the same age.

Note: Children's income is the sum of individual and spousal income at age 30, excluding immigrants after 1994. Parental income is the sum of the spouses' incomes for families in which the highest earner is ages 25-35.

Source: Chetty et al. (2017).

Figure 8: Percent of men in bottom or top earnings quartile with fathers in the bottom quartile



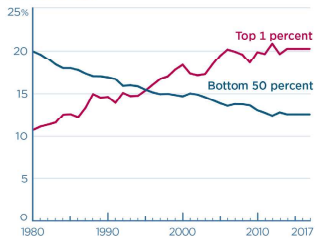
Americans born into low-income households are more likely to remain at the bottom of the income ladder than their European counterparts, according to an OECD report.

Note: Calculations based on 2011 data. Chile refers to 2009 data. Numbers shown on bars are rounded.

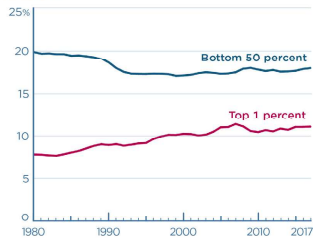
Source: Organization for Economic Cooperation and Development (OECD), "A Broken Social Elevator? How to Promote Social Mobility."

Figure 9: Pretax share of national income, 1980–2017

a. United States



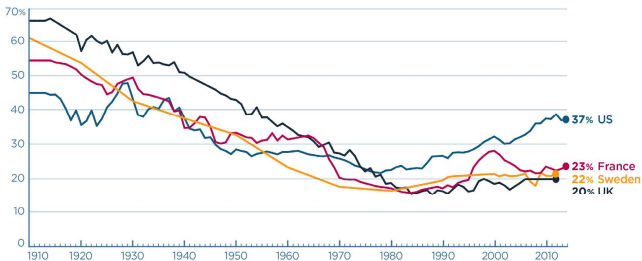
b. Europe



In the United States, the top 1 percent of earners now receive a larger pretax share of national income than their European equivalents.

Sources: Chancel (2019) based on Blanchet, Chancel, and Gethin (2019); Piketty, Saez, and Zucman (2018).

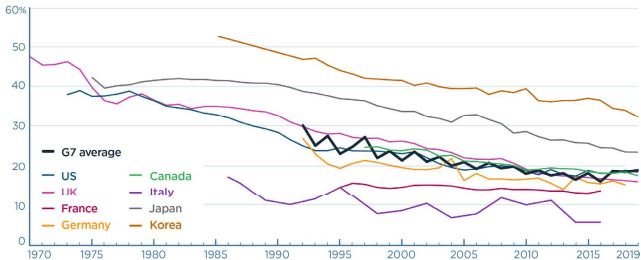
Figure 10: Top 1 percent personal wealth share in rich countries, 1910–2014



The wealthiest 1 percent of Americans have held close to 40 percent of the nation's wealth in recent years, approaching levels not seen in nearly a century.

Sources: Chancel (2019) based on Saez and Zucman (2016) for the United States; Garbinti, Goupille-Lebret, and Piketty (2018) for France; Alvaredo, Atkinson, and Morelli (2018) for the UK; Lundberg and Waldenström (2017) and Piketty (2014) for Sweden. See 2018 World Inequality Report for data.

Figure 11: Gender wage gap among G7 countries and South Korea, 1970–2019



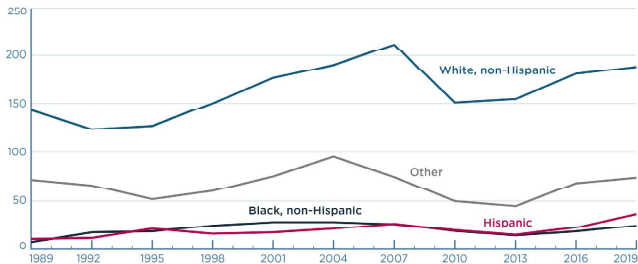
G7 = Group of Seven

Note: Shows difference between median earnings of men and women relative to median earnings of men.

Source: Organization for Economic Cooperation and Development (OECD).

Progress towards narrowing the gender wage gap slowed starting around 2005.

Figure 12: Median US household net worth by race, thousands USD

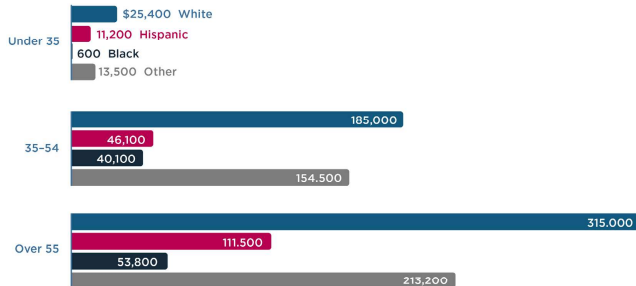


Wealth among Black and Hispanic families has grown more slowly since the 1990s than in white families.

Note: Household net worth is the sum of all financial and nonfinancial assets, minus debts, given in 2019 dollars.

Source: Federal Reserve Board, 2019 Survey of Consumer Finances.

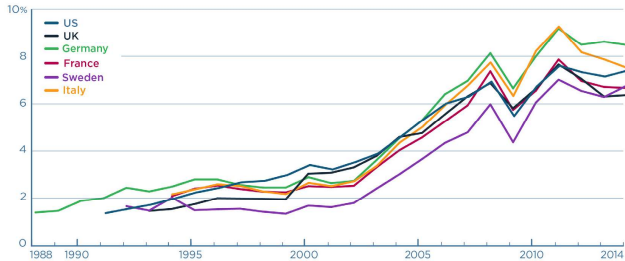
Figure 13: Median US wealth by age group and race/ethnicity, 2019, USD



People generally accumulate wealth as they grow older, but the differences in wealth across US racial and ethnic groups is stark. A 2019 Fed survey shows young Black families have almost no wealth (\$600) compared with \$25,400 for white families.

Source: Federal Reserve Board, 2019 Survey of Consumer Finances.

Figure 14: Imports from emerging-market countries as percent of GDP, 1988-2014

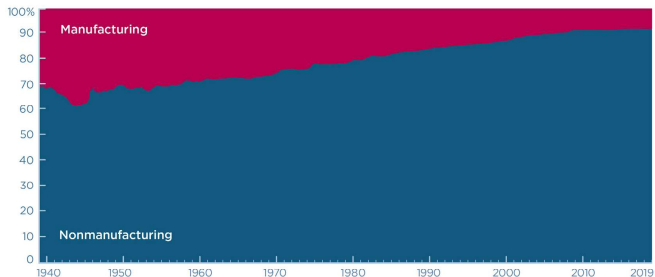


Note: Shows import penetration from low- and middle-income countries per World Bank classification.

Source: Chancel (2019) based on 2018 World Inequality Report.

Trade data show how the United States and many European countries have increased their imports from emerging-market countries at similar levels since 1980, suggesting factors other than trade are influencing differences in inequality levels.

Figure 15: Percent of US employment in manufacturing vs. nonmanufacturing industries, 1939–2019

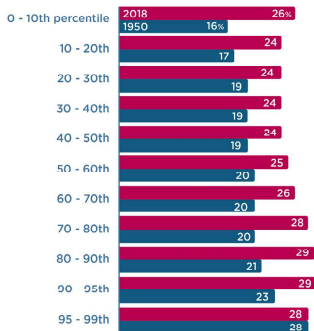


Note: Data exclude farming sector. Nonmanufacturing includes mining and logging, construction, private services, and government employment. Annual values are averages of monthly data.

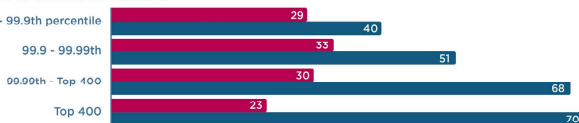
Source: US Bureau of Labor Statistics, Current Employment Statistics.

Manufacturing as a share of total employment has been in decline since the 1940s.

a. Bottom 99% of income distribution



b. Top 1% of income distribution



The average tax rate paid by the top 1 percent of US earners has steadily declined over many decades; since 2010, the highest-earning individuals have been paying an average tax rate roughly equal to or even less than other Americans. (Income levels of taxpayers do not account for government transfers).

Note: Tax rate includes all federal, state, and local taxes. Taxes are expressed as a fraction of pretax income. 0-10th percentile denotes the bottom 10 percent of the income distribution. Top 400 refers to the Forbes list of the 400 highest-earning Americans.

Source: Saez and Zucman (2019).

Figure 17: Top marginal income tax rate, 1900-2020

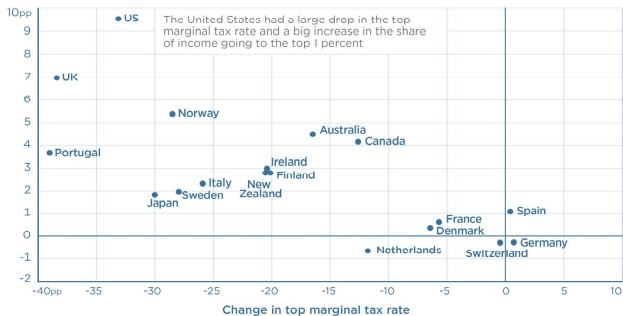


In the 1950s, the top US marginal income tax rate was above 90 percent, a legacy of the war-footing economy of World War II. It is now just below 40 percent. (In the United States, the top marginal tax rate is charged only on earnings above \$510,000 for an individual). Other countries, like Japan and France, had similar declines.

Source: Chancel (2019) based on Piketty (2014) and updates; see 2018 World Inequality Report for data; Organisation for Economic Co-operation and Development (2018-2020).

Figure 18: Change in top marginal tax rate vs. change in share of income held by top 1 percent since the 1970s

Change in top 1 percent income share

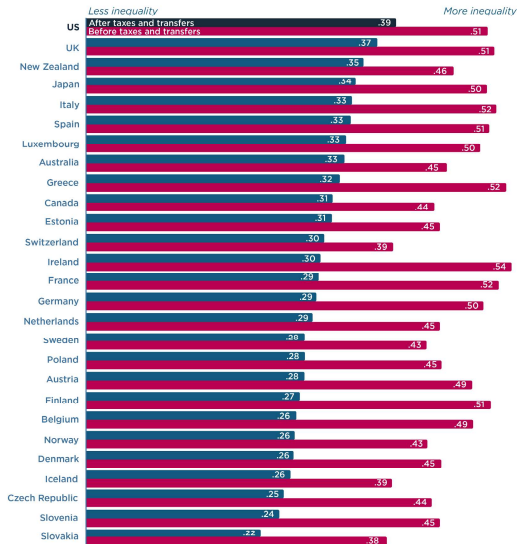


pp = percentage points

As top marginal tax rates have declined in many richer countries, the share of income going to the top 1 percent of earners has grown. Countries like Germany, Spain, Denmark, and Switzerland cut top tax rates by little or not at all, and also experienced little to no increase in top income shares.

Sources: Chancel (2019) based on Piketty, Saez, and Stantcheva (2014); see 2018 World Inequality Report for data.

Figure 20: Gini coefficient before and after taxes and transfers for high-income OECD countries, 2018 or latest available data



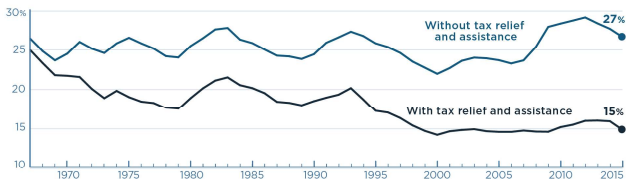
Note: The Gini coefficient measures how equally income is distributed across a population, from 0 (perfectly equal) to 1 (all income to one person). Direct taxes exclude sales and other indirect taxes. Data years range from 2014–2019.

Sources: Organization for Economic Cooperation and Development (OECD). Calculations by Jacob Funk Kirkegaard and David Xu.

Before direct taxes and government transfers, US inequality (measured by the Gini coefficient) is similar to other wealthy countries. But after tax and transfers take effect, it ranks the highest, indicating the US government is doing less than other governments to mitigate inequality.

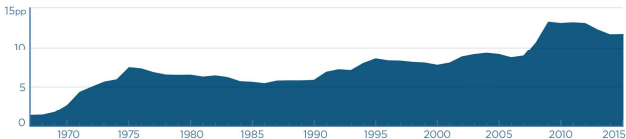
Figure 21

a. US poverty levels with and without tax relief and government assistance programs, 1967–2015



Since 1965, the gradual expansion of the social safety net has brought post-tax-and-transfer poverty, the poverty level after government tax relief and assistance programs are administered, down by around 10 percentage points.

b. Difference between poverty levels with and without tax relief and government assistance, 1967–2015



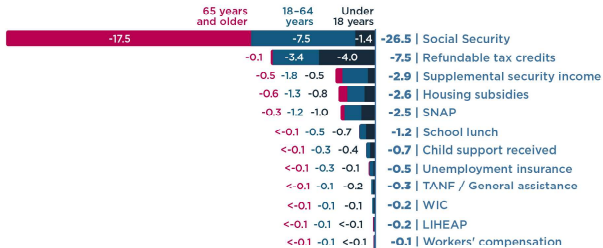
pp = percentage points

Note: Refers to market income poverty and the supplemental poverty measure (post-tax-and-transfer), which adds cash transfers and the cash equivalent of in-kind transfers to family earnings, minus net tax liabilities. The supplemental poverty measure also indirectly incorporates health benefits by subtracting out-of-pocket health costs from income.

Source: Based on Wimer et al. (2013), updated November 2016 as reported in Furman (2017).

Figure 22: Change in number of people in poverty in the US after government transfers and nondiscretionary expenses by age group, millions, 2019

a. Elements that decreased poverty



b. Elements that increased poverty



The US government transfer that does the most in reducing poverty is Social Security followed by refundable tax credits.

SNAP = Supplemental Nutrition Assistance Program; TANF = Temporary Assistance for Needy Families; WIC = The Special Supplemental Nutrition Program for Women, Infants, and Children; LIHEAP = Low Income Home Energy Assistance Program; FICA = Federal Insurance Contributions Act

Source: US Census Bureau, Current Population Survey, 2020 Annual Social and Economic Supplement.

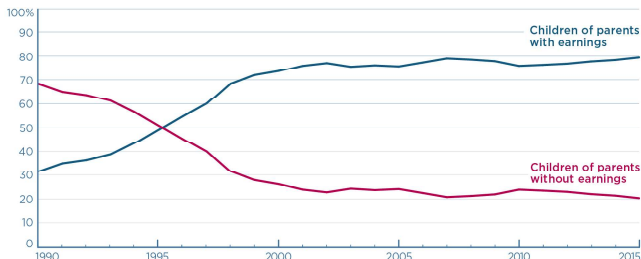
Figure 23: Ratio of wages earned by college graduates vs. high school graduates, 1914–2017



Source: Autor, Goldin, and Katz (2020).

In the last 40 years, a college degree has become the “big dividing line” as wages for college graduates have increased at a much faster rate than high school graduates.

Figure 24: Share of federal spending on children by parental earnings, 1990–2015



Source: Hoynes and Schanzenbach (2018).

In the 1990s, social safety net spending was targeted at families without earnings. By the 2000s, the trend had reversed, and the federal government began spending more on families with earnings than on those without.

What income group is affected?	At what stage of the economy does the policy intervene?		
	Pre-production	Production	Post-production
Bottom earners	<i>Expanded access to healthcare and education; universal basic income</i>	<i>Minimum wage policies; job guarantees</i>	<i>Government transfers, including Earned Income Tax Credit (EITC); full employment policies</i>
Middle earners	<i>Public spending on higher education and job training programs</i>	<i>"Good jobs" policies; industrial relations and labor laws; sectoral wage boards; trade agreements; innovation policies</i>	<i>Social safety nets; social insurance policies</i>
Top earners	<i>Inheritance and estate taxes</i>	<i>Corporate regulation; antitrust laws</i>	<i>Wealth taxes</i>

Source: Olivier Blanchard and Dani Rodrik, PIIE blog post, "We Have the Tools to Reverse the Rise in Inequality."

Another way of thinking about policies to combat inequality is to organize them by economic stages: Policies in the pre-production stage focus on people entering the workforce, production policies affect workers, and post-production policies redistribute income and wealth. These categories can then be further divided by policies directed at bottom, middle, and top income levels. Listen to Dani Rodrik explain here.

Covid-19 and inequalities

Far from pushing inequality down the agenda, the pandemic has reinforced the need to deal with the challenges posed by inequality

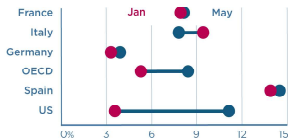
- Highlighted existing inequalities
 - Savings and debt
 - Housing
 - State of benefit system
- Exacerbated some inequalities
 - By income
 - Intergenerational
 - Gender
 - Ethnicity



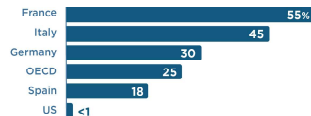
This presentation

Figure 19

**a. Unemployment rates,
January and May, 2020**



**b. Share of employees who claimed job
retention support, January-May, 2020**



Note: US unemployment data refer to June and include temporary layoffs. US job retention support data refer to short time compensation schemes.

Source: Organization for Economic Cooperation and Development (OECD), Employment Outlook 2020.

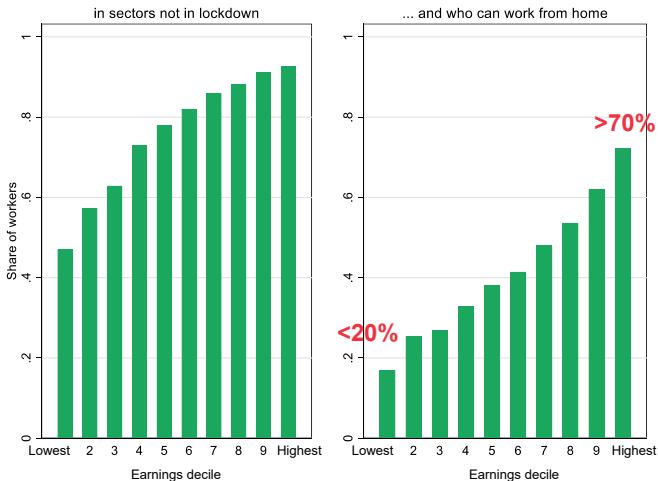
European countries like France, Italy, and Germany were better able to keep workers employed during the pandemic than the United States through existing “short-time” work schemes and wage subsidies to businesses.

Sector shutdowns and home-working have hit low-paid workers the hardest



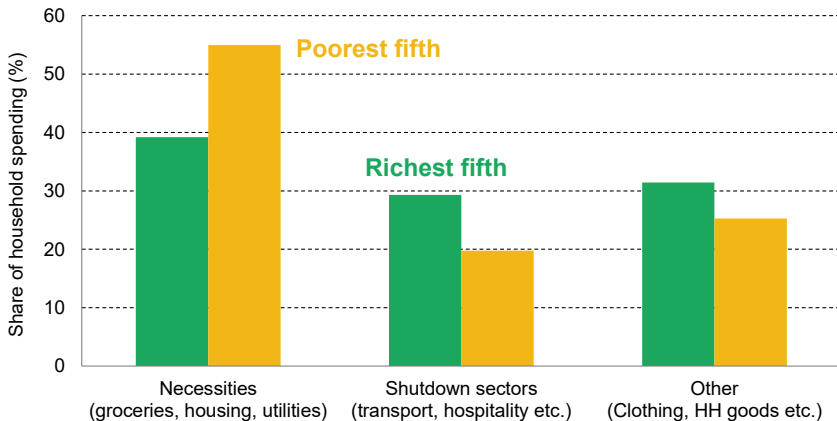
Inequality

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Source: Blundell et al. 2020

Whilst high-income families may be forced to save



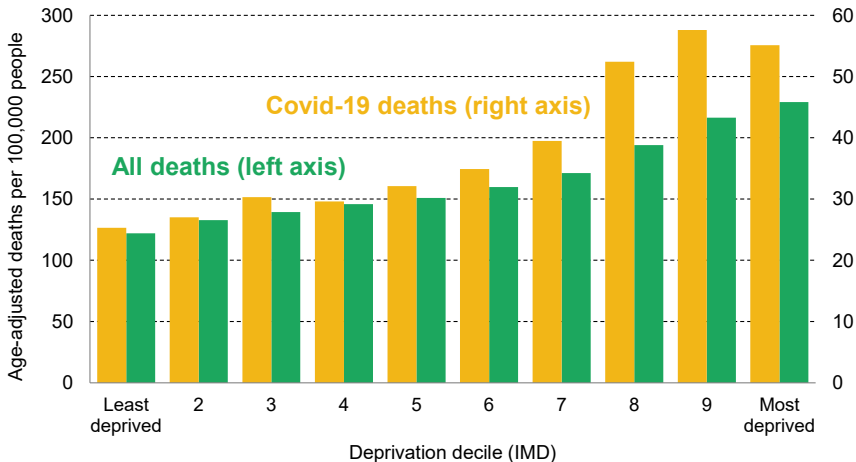
Source: Crawford et al. 2020



Inequality

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Covid-19 death rates are much higher in more deprived areas



Source: ONS 2020

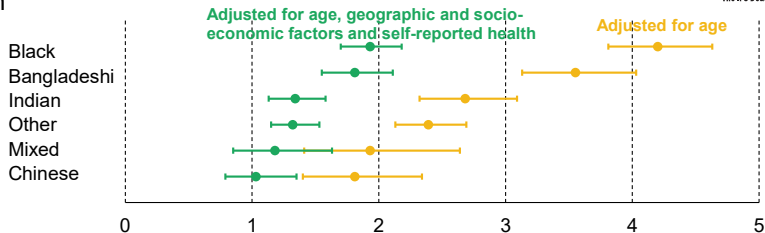
Ethnic minorities have higher Covid-19 death rates even adjusting for demographics



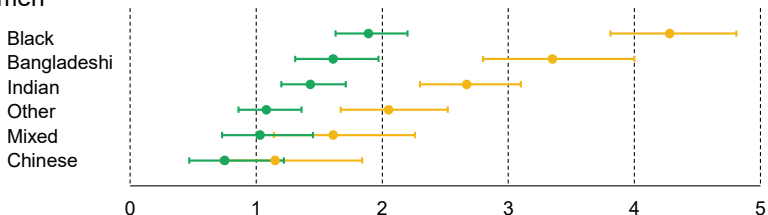
Inequality

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Men



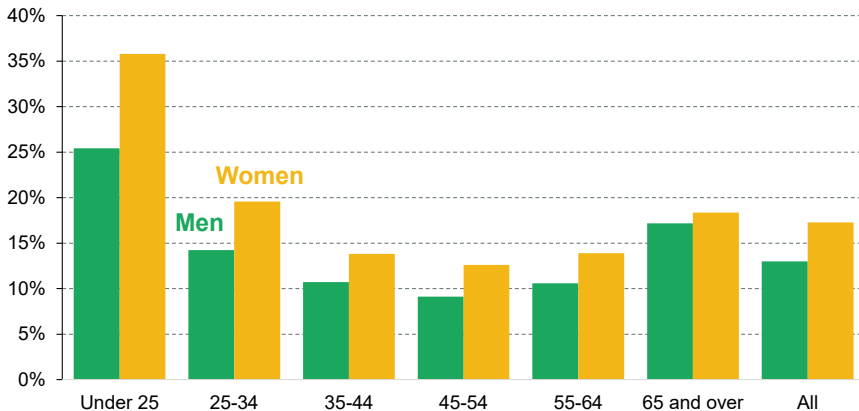
Women



Source: ONS 2020

Likelihood of dying from COVID-19 compared to white ethnicity

Young people are much more likely to work in shutdown sectors



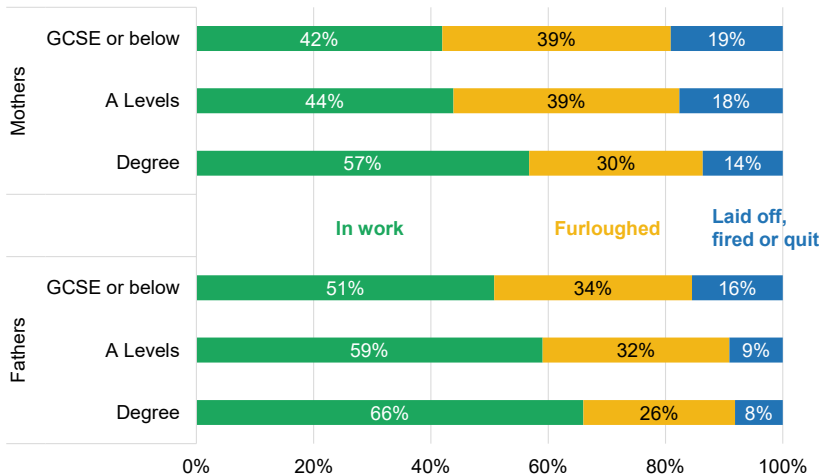
Source: Joyce and Xu 2020



Inequality

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Mothers are more likely to have had their careers disrupted



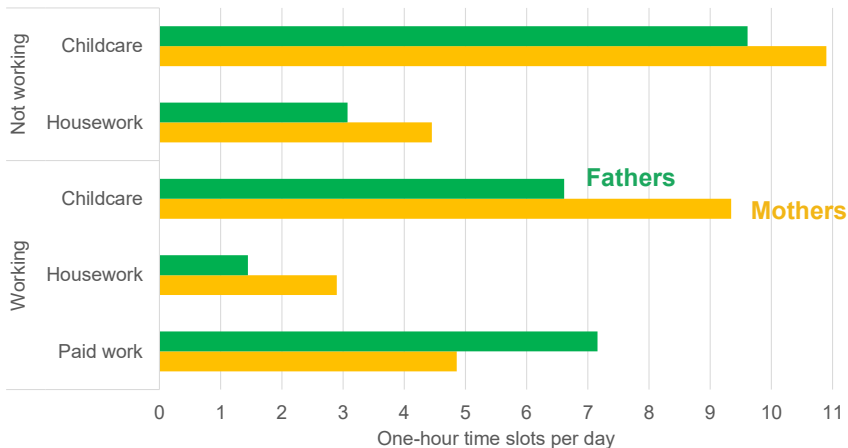
Source: Andrew et al. 2020



Inequality

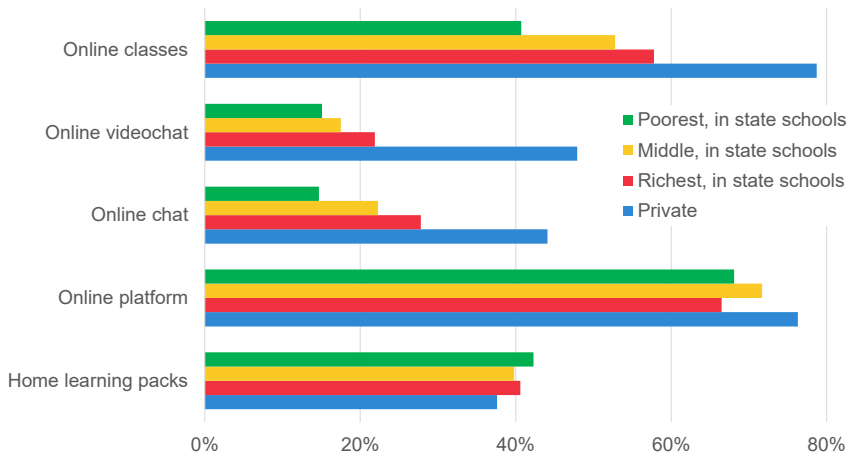
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And bear most of the additional childcare



Source: Andrew et al. 2020

School closures are likely to exacerbate educational inequalities



Source: Andrew et al. 2020

Discussion on Readings and COVID-19

1. What are the key trade-offs between economic activity and health outcomes in the COVID-19 pandemic?
2. Macroeconomic Activity and the Pandemic. How does the current recession differ from the 2008-2009 economic crisis?
3. Macroeconomic Policy Response. Key channels of policy's impact.
4. The shape of the economic recovery. What does it depend on? What are the potential consequences of excess savings of the household sector?
5. Biden's gamble. Hopes and fears around the U.S. fiscal stimulus.
6. Your thoughts on inequality in the post-COVID world.

Final Exam Details

- ▶ **Date: April 21st, 5pm - 7pm, Online Synchronous, Zoom Link will be posted on Quercus.**
- ▶ Additional 30 minutes to submit the solutions in a **single PDF file**.
- ▶ Non-cumulative and open book. Coverage: Chapter 14, Chapter 18, Financial Crisis and COVID-19 lectures. The format will be identical to the term tests.
- ▶ Best preparation: go over the Practice Final Exam, Tutorial Questions and end-of-chapter Problems.
- ▶ Grades will be curved. **I want you to do well in the course!**

Course Evaluations Are Now Open!

You will now have time to provide feedback on this course

Bring any of the following electronic devices to class:

- Smart phone
- Tablet
- Laptop



Check your e-mail for a link to your evaluations, or go to

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